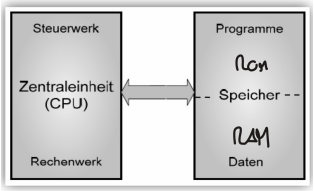
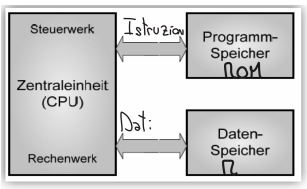
Quick Reference

Lars Kamm, Jan Wendler



Von Neumann Architektur

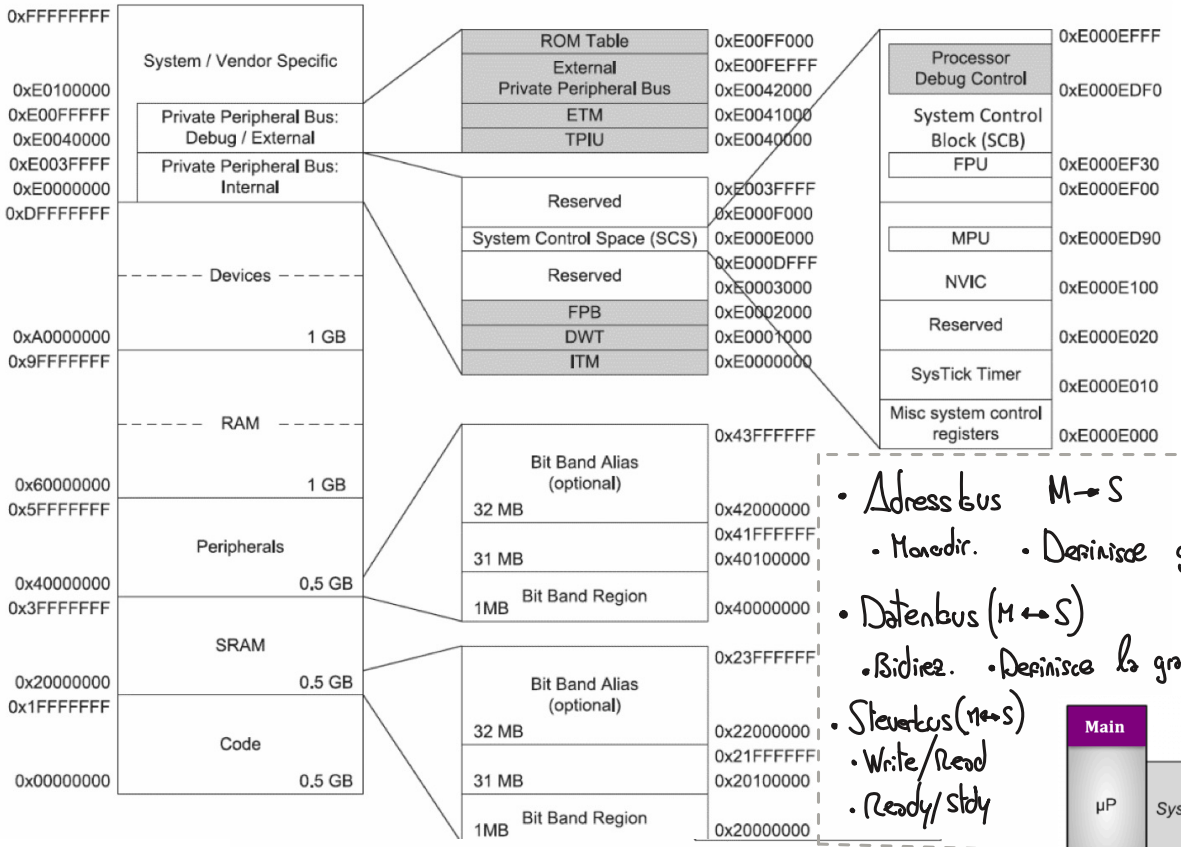


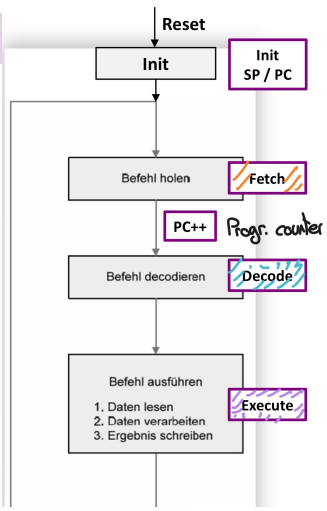
Harvard Architektur

- La grandezza della ALU definisce la grandezza di una Word
- RISC: Pochi comandi

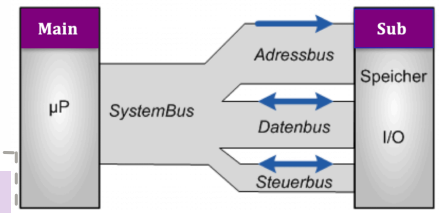
1 ARM CORTEX-M MEMORY ORGANIZATION

1.1 ARM Cortex-M Memory-Map





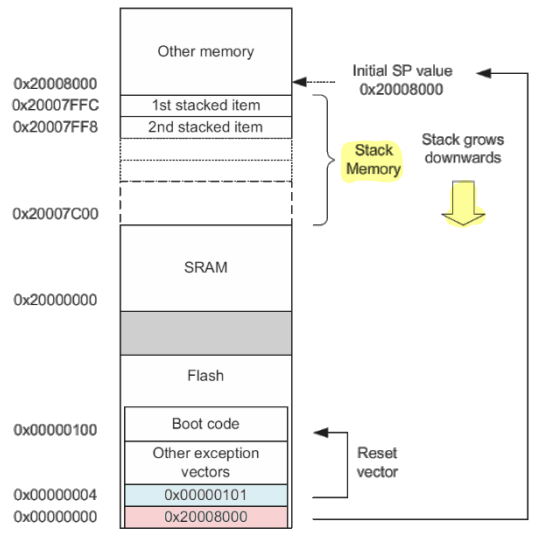
- Adressbus M → S
  - Monodir.
  - Definisce grandezza Adressraum
- Datenbus (M ↔ S)
  - Bidirez.
  - Definisce la grandezza di trasferimento dati
- Steuerbus (M ↔ S)
  - Write/Read
  - Ready/Stby



1.2 Vector-Table & Reset-Sequence (Processo di avvio)

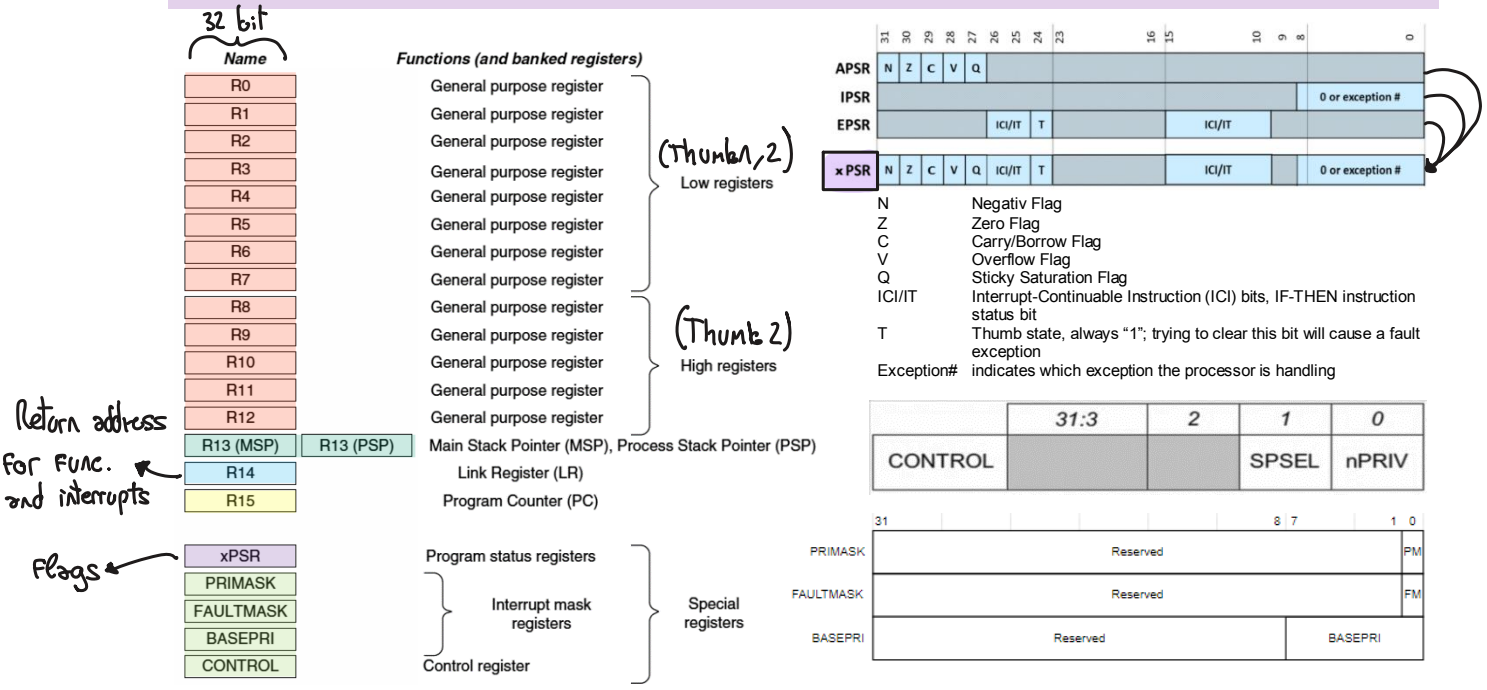
Exception Type	CMSIS Interrupt Number	Address Offset	Vectors
18 - 255	2 - 239	0x48 – 0x3FF	IRQ #2 - #239 [1]
17	1	0x44	IRQ #1 [1]
16	0	0x40	IRQ #0 [1]
15	-1	0x3C	SysTick [1]
14	-2	0x38	PendSV [1]
NA	NA	0x34	Reserved
12	-4	0x30	Debug Monitor [1]
11	-5	0x2C	SVC [1]
NA	NA	0x28	Reserved
NA	NA	0x24	Reserved
NA	NA	0x20	Reserved
NA	NA	0x1C	Reserved
6	-10	0x18	Usage fault [1]
4	-11	0x14	Bus Fault [1]
4	-12	0x10	MemManage Fault [1]
3	-13	0x0C	HardFault [1]
2	-14	0x08	NMI [1]
1	NA	0x04	Reset [1]
NA	NA	0x00	Initial value of MSP

Anomalie termiche Watchdog timer



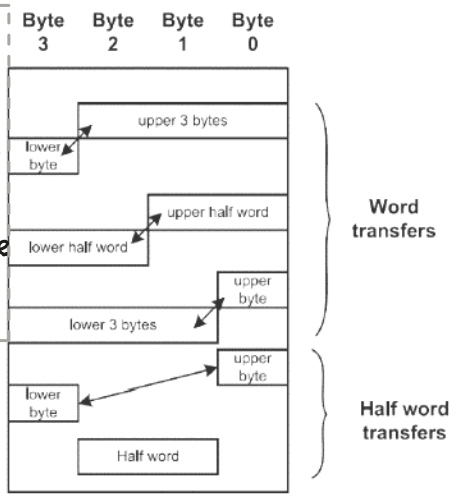
Set Program Ctr. (Indirizzo prima istruzione)
Set Stack ptr. (Dove inizia la RAM dello stack)

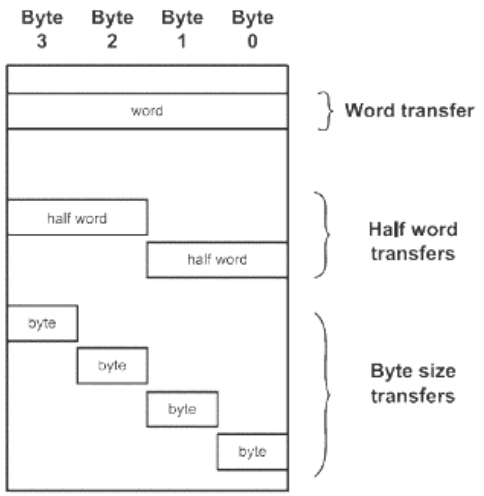
1.3 Register-Set & Program Status Register



1.4 Alignement (Cortex-M ⇒ Bus indirizzi di 32 bit (4 byte))

● : Il processore deve eseguire ⊕ dato zugriff leggendo solo specifiche porzioni del bus e rimappandole correttamente



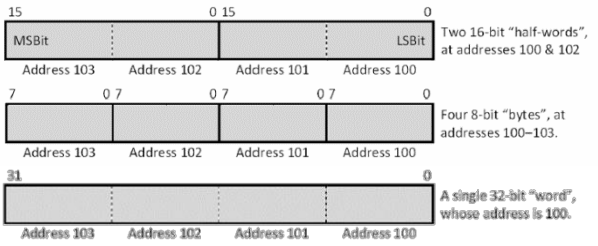


Cortex-M

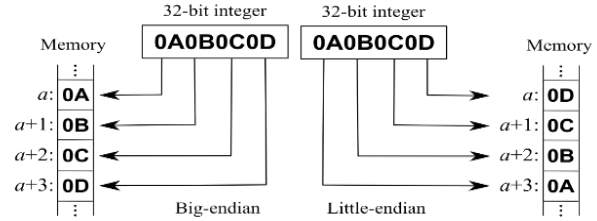
Unaligned transfers

Word (32Bit):	0x0,	0x4,	0x8	0xC
	Accesso solo a multipli di 4			
Halfword (16Bit):	0x0,	0x2,	0x4,	0x6
	0x8,	0xA,	0xC,	0xE
Byte (8Bit):	0x0,	0x1,	….,	0xF

Aligned transfers



1.5 Endianess (Cortex-M ⇒ Little-endian)



Endianess betrifft nur Zahlenformate die grösser als ein Byte sind (Halfword, Word, Doubleword).

Little-Endian: Least Significant Byte befindet sich an der tiefsten Adresse.
⊕ Usato

Big-Endian: Most Significant Byte befindet sich an der tiefsten Adresse.

Un indirizzo in memoria punta SEMPRE al byte più basso della var. La var. si estende SEMPRE verso indir. ⊕ Alti

1.6 Speichergrössen

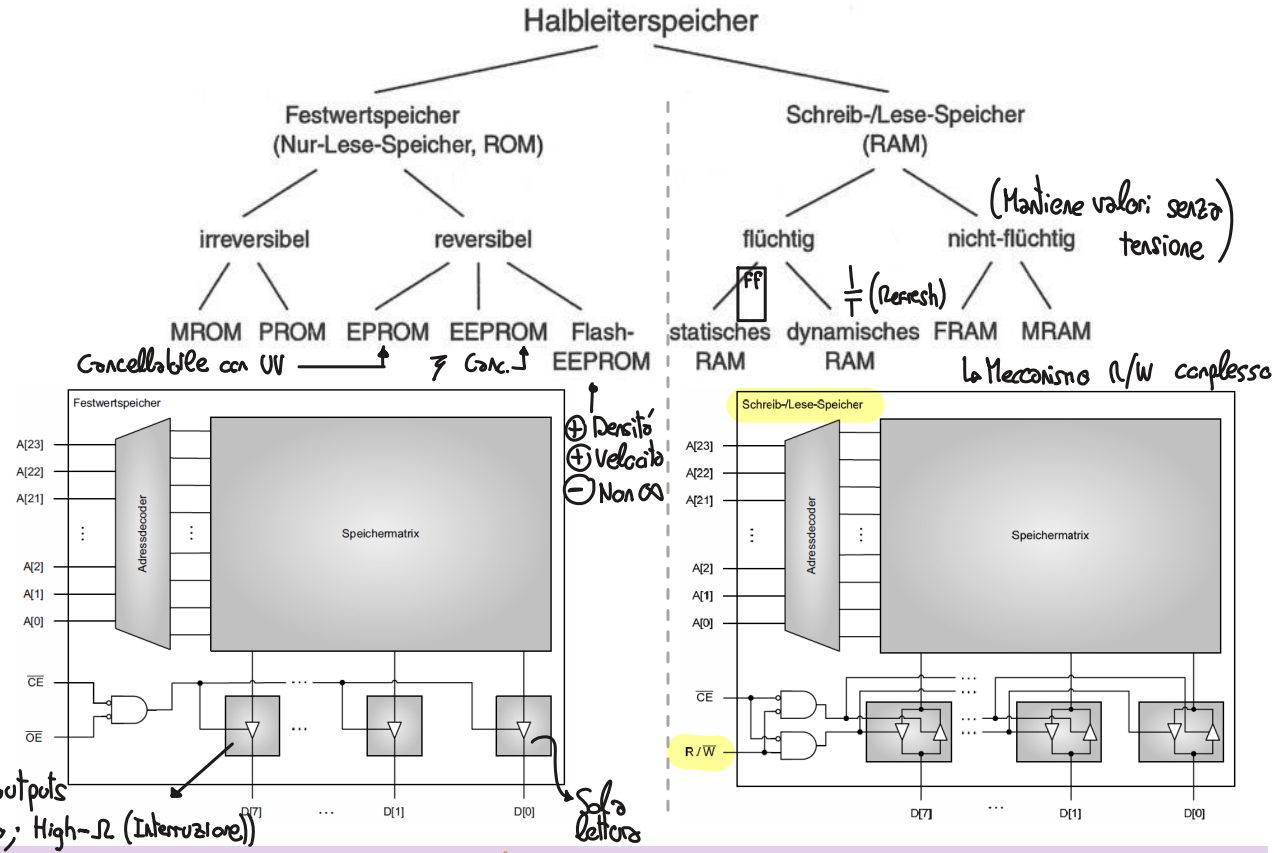
Zur Angabe der Speicherkapazität als Anzahl von Bits, Bytes oder Codewörtern verwendet man in der Informatik in Anlehnung an die Physik die Bezeichnungen Kilo (K), Mega (M), Giga (G) und Tera (T). Diese beziehen sich auf das Binärzahlensystem:

$$\mathbf{1K} = 2^{10} = \mathbf{1'024}$$
$$\mathbf{1M} = 2^{20} = \mathbf{1'024K} = \mathbf{1'048'576}$$
$$\mathbf{1G} = 2^{30} = \mathbf{1'024M} = \mathbf{1'048'576K} = \mathbf{1'073'741'824}$$
$$\mathbf{1T} = 2^{40} = \mathbf{1'024G} = \mathbf{1'048'576M} = \mathbf{1'073'741'824K} = \mathbf{1'099'511'627'776}$$

Memory mapped
↳ Gemeinsamer Adressraum für I/O und Speicher

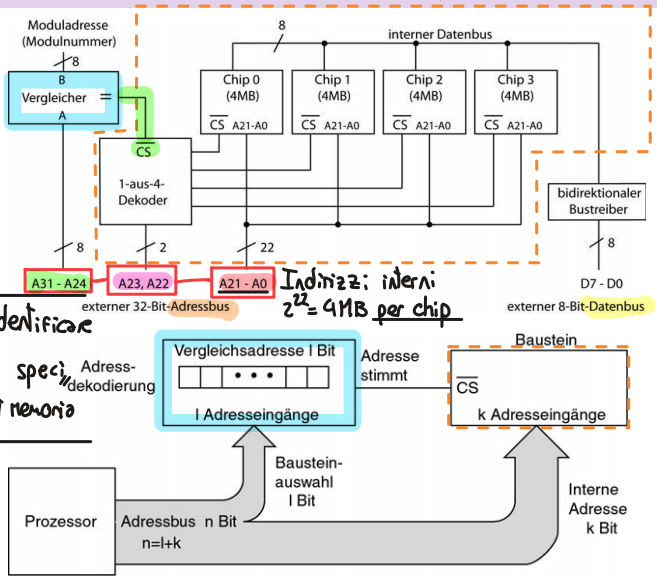
Isolierte Addressierung
↳ Separater Adressraum für I/O und Speicher

1.7 Speicherarten



1.8 Adressdekodierung

Modulo di memoria



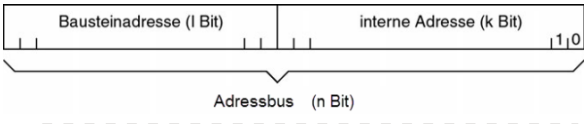
$l = 8$

Servono ad identificare se attivare lo specifico Blocco di memoria

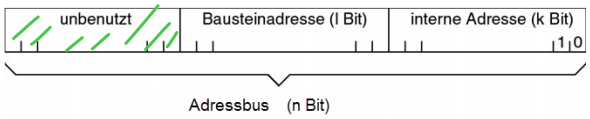
● = 2 Bit aggiuntivi per selezionare il chip (Decoder one-Cold)

Esempio $(22+2) + 8 = 32$

$$k + l = n$$



$$k + l < n$$

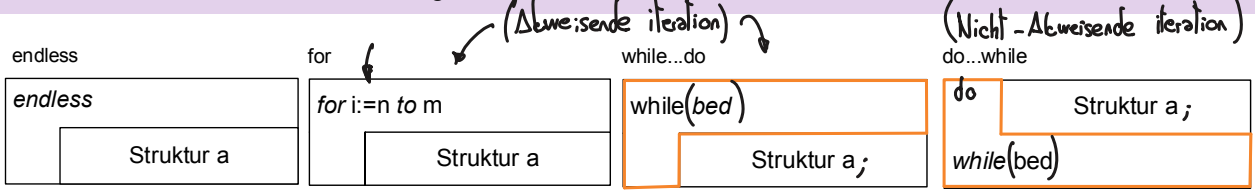


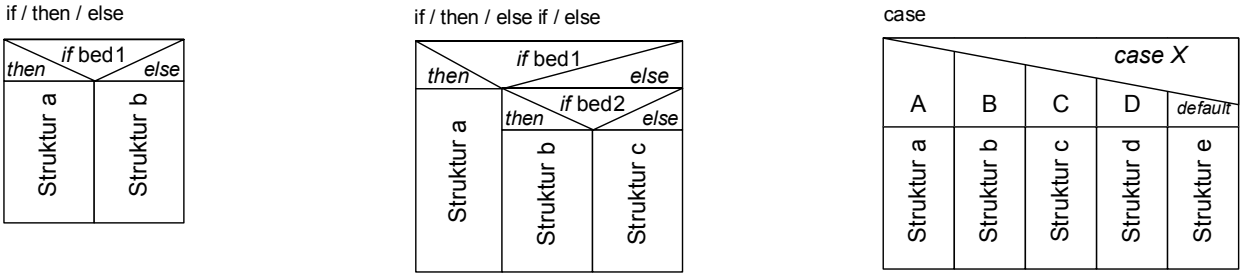
(Chip = Contenitore di memoria)

CS = Chip select (Accensione chip)

2 KONTROLL- & SELEKTIONSSTRUKTUREN

2.1 Nassi Schneidermann Diagramme

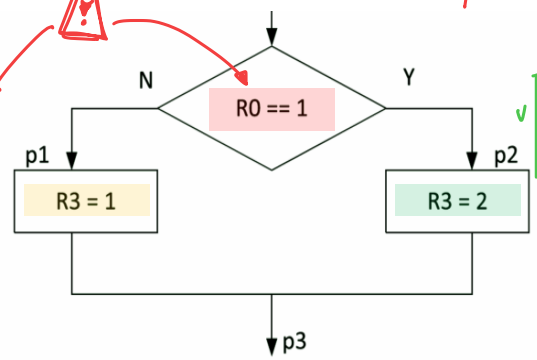




2.2 Entscheidungen treffen

Le condizioni in assembly sono invertite (ordine istruzioni mantenuto ☺)

```
// a → R0
// b → R3
if(a != 1)
{
  b = 1;
}
else
{
  b = 2;
}
```




```
  CMP  r0, #1 ;compare r0 to 1
  BEQ  p2     ;if equal, go to p2
  MOVS r3, #1 ;then r3 = 1
  B    p3     ;go to end (p3)
p2:           ;label p2
  MOVS r3, #2 ;else r3 = 2
p3:           ;label p3 (else)
  …           ;other operations
```

In Assembler-Sprache ist eine Entscheidung praktisch immer ein 2-stufiger Ablauf

1. **Benötigte Flags ermitteln**
   Vergleich (**Compare**) vornehmen, indem zwei Werte voneinander subtrahiert werden. Bei der Differenzbildung ist das Resultat nicht wichtig, sondern nur dessen Eigenschaften (**Flags**).

2. **Zugehörige Sprünge ausführen**
   Im zweiten Schritt werden mit den Flags bedingte Sprünge (**B{cond}**) ausgeführt

2.3 Umsetzung von C/C++ Strukturen

In questo esempio non viene eseguita l'inversione (caso speciale)

Thumb-2 Assembly	C Source Code
…	uint32_t a; // unsigned 32Bit
while1:	…
LDR R0, =a	while(1) // endless loop?
LDR R0, [R0]	{
CMP R0, #0x05	if(a == 5)
BEQ end_while1	{
BL foo	break;
…	}
B while1	foo();
end_while1:	…
…	}
	…

Ptr in memoria

Normalmente viene invertita la cond. per "saltare" subito. visto che ora tramite break bisogna saltare proprio se 'if'

nei loop la condizione non è invertita

3 ARM CORTEX-M0(+) INSTRUCTION-SET

3.1 Symbole & Appendix

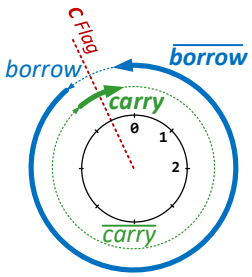
Symbols	Meaning
R_d R_m R_n	represent 32-bit registers
{R_d,}	if R_d is present R_d is destination, otherwise R_n
{S}	if S is present, instruction will set condition codes (Aggiornamento flags)
{cc}	optional logical condition, condition code suffix (EQ, NE, GT, LT, HI) (= ≠ > <)
#imm3	any value in the range: 0...7
#imm5	any value in the range: 0...31
#imm7	any value in the range: 0..127
#imm8	any value in the range: 0..255
#imm10	any value in the range: 0..1023
#imm11	any value in the range: 0..2047
label	any address within the ROM of the microcontroller; offset range relative to PC: B *label* −2 KB to +2 KB B<cc> *label* −256 bytes to +255 bytes BL *label* −16 MB to +16 MB (32bit Befehl)
{reglist}	List of registers, sequence is not relevant (Ogni bit della lista indica un registro … 0 0 1 1 0 0 0 R8 R5 R4 R3 R2 R1 R0)

Parentesi graffe : opzionali

3.2 Bedingte Ausführung & Code-Zusätze {cc}

(Branches)

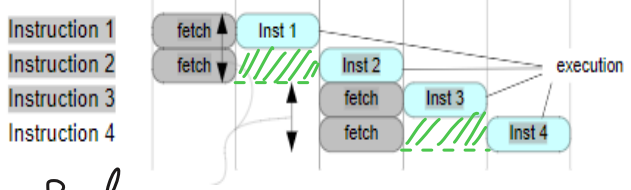
cc	Meaning	Flags (APSR) / Requirements	Binary	uint		int	
EQ	Equal	Z = 1	0000	X	==	X	==
NE	Not equal	Z = 0	0001	X	!=	X	!=
CS or **HS**	Carry set, Unsigned ≥ carry OR $\overline{\text{borrow}}$	C = 1	0010	X	>=		
CC or **LO**	Carry clear,Unsigned < $\overline{\text{carry}}$ OR borrow	C = 0	0011	X	<		
MI	Minus/negative	N = 1	0100			X	-
PL	Positive or zero (non-negative)	N = 0	0101			X	+
VS	Overflow	V = 1	0110			X	
VC	No overflow	V = 0	0111			X	
HI	Unsigned > ("Higher")	C = 1 && Z = 0	1000	X	>		
LS	Unsigned ≤ ("Lower or Same")	C = 0 || Z=1	1001	X	<=		
GE	Signed ≥ ("Greater than or Equal")	N = V	1010			X	>=
LT	Signed < ("Less Than")	N ≠ V	1011			X	<
GT	Signed > ("Greater Than")	Z = 0 && N = V	1100			X	>
LE	Signed ≤ ("Less than or Equal")	Z = 1 || N ≠ V	1101			X	<=
AL	Always (unconditional)	any	1110				



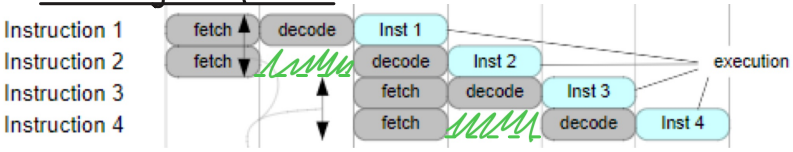
ACHTUNG:
CMP ist eine Subtraktion (Borrow)
CMN ist eine Addition (Carry)

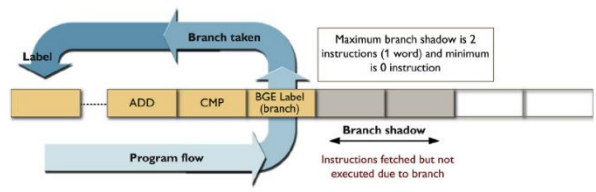
3.3 Pipelining

2-Stage Pipeline



3-Stage Pipeline



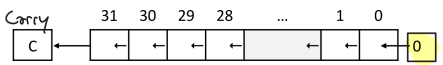


Bei Sprüngen muss die Pipeline "geleert" werden

Systembus frei

(? → Flag modificata)

logical shift left

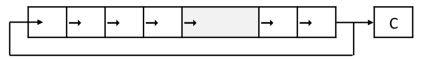


logical shift right

Arithmetic Shift right

(Vorz. mantenuto Per signed var.)

Rotate Shift right



Group	Description		Syntax	Cycles	Flags				Instruction Code 16Bit																
					N	Z	C	V	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Shift (immediate), add, subtract, move, and compare	Move Lo to Lo	$Rd_{0-7} \leftarrow Rm_{0-7}$	MOVS Rd, Rm	1	?	?	-	-	0	0	0	0	0	0	0	0	0	0	Rm			Rd		
	Logical shift left by immediate	$Rd_{0-7} \leftarrow Rm_{0-7}$ << #imm5	LSLS Rd, Rm, #<imm>	1	?	?	?	-	0	0	0	0	0	imm5			Rm			Rd				
	Logical shift right by immediate	$Rd_{0-7} \leftarrow Rm_{0-7}$ >> #imm5	LSRS Rd, Rm, #<imm>	1	?	?	?	-	0	0	0	0	1	imm5			Rm			Rd				
	Arithmetic shift right	$Rd_{0-7} \leftarrow Rm_{0-7}$ >> #imm5	ASRS Rd, Rm, #<imm>	1	?	?	?	-	0	0	0	1	0	imm5			Rm			Rd				
	Add 3-bit immediate	$Rd_{0-7} \leftarrow Rm_{0-7}$ + #imm3	ADDS Rd, Rm, #<imm>	1	?	?	?	?	0	0	0	1	1	1	0	imm3		Rm			Rd				
	Add All registers Lo	$Rd_{0-7} \leftarrow Rn_{0-7} + Rm_{0-7}$	ADDS Rd, Rn, Rm	1	?	?	?	?	0	0	0	1	1	0	0	Rm		Rn			Rd				
	Sub Lo and Lo	$Rd_{0-7} \leftarrow Rn_{0-7} - Rm_{0-7}$	SUBS Rd, Rn, Rm	1	?	?	?	?	0	0	0	1	1	0	1	Rm		Rn			Rd				
	Sub 3-bit immediate	$Rd_{0-7} \leftarrow Rn_{0-7}$ - #imm3	SUBS Rd, Rn, #<imm>	1	?	?	?	?	0	0	0	1	1	1	1	imm3		Rn			Rd				
	Move 8-bit immediate	$Rd_{0-7} \leftarrow$ #imm8	MOVS Rd, #<imm>	1	?	?	-	-	0	0	1	0	0	Rd			imm8								
	Compare Immediate	Rn_{0-7} - #imm8 == 0	CMP Rn, #<imm>	1	?	?	?	?	0	0	1	0	1	Rn			imm8								
	Add 8-bit immediate	$Rd_{0-7} \leftarrow Rd_{0-7}$ + #imm8	ADDS Rd, #<imm>	1	?	?	?	?	0	0	1	1	0	Rd			imm8								
	Sub 8-bit immediate	$Rd_{0-7} \leftarrow Rd_{0-7}$ - #imm8	SUBS Rd, #<imm>	1	?	?	?	?	0	0	1	1	1	Rd			imm8								
Data processing	AND	$Rd_{0-7} \leftarrow Rd_{0-7}$ & Rm_{0-7}	ANDS Rd, Rm	1	?	?	-	-	0	1	0	0	0	0	0	0	0	0	Rm			Rd		
	Exclusive OR	$Rd_{0-7} \leftarrow Rd_{0-7}$ ^ Rm_{0-7}	EORS Rd, Rm	1	?	?	-	-	0	1	0	0	0	0	0	0	0	1	Rm			Rd		
	Logical shift left by register	$Rd_{0-7} \leftarrow Rd_{0-7}$ << Rm_{0-7}	LSLS Rd, Rm	1	?	?	?	-	0	1	0	0	0	0	0	0	1	0	Rm			Rd		
	Logical shift right by register	$Rd_{0-7} \leftarrow Rd_{0-7}$ >> Rm_{0-7}	LSRS Rd, Rm	1	?	?	?	-	0	1	0	0	0	0	0	0	1	1	Rm			Rd		
	Arithmetic shift right by register	$Rd_{0-7} \leftarrow Rd_{0-7}$ >> Rm_{0-7}	ASRS Rd, Rm	1	?	?	?	-	0	1	0	0	0	0	0	1	0	0	Rm			Rd		
	Add with carry	$Rd_{0-7} \leftarrow Rd_{0-7} + Rm_{0-7}$	ADCS Rd, Rm	1	?	?	?	?	0	1	0	0	0	0	0	1	0	1	Rm			Rd		
	Sub with carry	$Rd_{0-7} \leftarrow Rd_{0-7}$ - Rm_{0-7}	SBCS Rd, Rm	1	?	?	?	?	0	1	0	0	0	0	0	1	1	0	Rm			Rd		
	Rotate right by register	$Rd_{0-7} \leftarrow Rd_{0-7}$ >>> Rm_{0-7}	RORS Rd, Rm	1	?	?	?	-	0	1	0	0	0	0	0	1	1	1	Rm			Rd		
	AND test (ANDS solo flags)	Rn_{0-7} & Rm_{0-7}	TST Rn, Rm	1	?	?	?	-	0	1	0	0	0	0	0	1	0	0	Rm			Rn		
	Negate	$Rd_{0-7} \leftarrow$ #0 -Rm	RSBS Rd, Rm, #0	1	?	?	?	?	0	1	0	0	0	0	0	1	0	1	Rm			Rd		
	Compare Lo to Lo	Rn_{0-7} - Rm_{0-7} == 0	CMP Rn, Rm	1	?	?	?	?	0	1	0	0	0	0	0	1	0	0	Rm			Rn		
	Compare Negative	Rn_{0-7} + Rm_{0-7} == 0	CMN Rn, Rm	1	?	?	?	?	0	1	0	0	0	0	0	1	0	1	Rm			Rn		
	OR	$Rd_{0-7} \leftarrow Rd_{0-7}$ | Rm_{0-7}	ORRS Rd, Rm	1	?	?	-	-	0	1	0	0	0	0	0	1	1	0	Rm			Rd		
	Multiply	$Rd_{0-7} \leftarrow (Rm_{0-7} * Rd_{0-7}) >> 32$	MULS Rd, Rm	1 or 32 (1)	?	?	-	-	0	1	0	0	0	0	0	1	1	0	Rm			Rd		
	Bit clear	$Rd_{0-7} \leftarrow Rd_{0-7}$ & ~(Rm_{0-7})	BICS Rd, Rm	1	?	?	-	-	0	1	0	0	0	0	0	1	1	1	Rm			Rd		
	Move NOT	$Rd_{0-7} \leftarrow$ ~(Rm_{0-7})	MVNS Rd, Rm	1	?	?	-	-	0	1	0	0	0	0	0	1	1	1	Rm			Rd		
Special data instructions and branch and exchange	Add Any to PC	$PC \leftarrow PC + Rm_{0-15}$	ADD PC, Rm	3	-	-	-	-	0	1	0	0	0	1	0	0	1		Rm			1	1	1
	Add Any to Any	$Rd_{0-15} \leftarrow Rd_{0-15} + Rm_{0-15}$	ADD Rd, Rm (ADD Rd,Rd,Rm)	1	-	-	-	-	0	1	0	0	0	1	0	0	Rd		Rm			Rd		
	Add address from SP and register	$Rd_{0-15} \leftarrow SP + Rd_{0-15}$	ADD Rd, SP	1	-	-	-	-	0	1	0	0	0	1	0	0	Rd	1	1	0	1	Rd		
	Compare Any to Any	Rn_{0-15} - Rm_{0-15} == 0	CMP Rn, Rm	1	?	?	?	?	0	1	0	0	0	1	0	1	Rn		Rm			Rn		
	Move Any to Any	$Rd_{0-15} \leftarrow Rm_{0-15}$	MOV Rd, Rm	1	-	-	-	-	0	1	0	0	0	1	1	0	Rd		Rm			Rd		
	Move Any to PC	$PC \leftarrow Rm_{0-15}$	MOV PC, Rm	2	-	-	-	-	0	1	0	0	0	1	1	0	1		Rm			1	1	1
	Branch With exchange	$PC \leftarrow R_{m0-15}$	BX Rm	2	-	-	-	-	0	1	0	0	0	1	1	1	0		Rm			0	0	0
	Branch With link and exchange	$PC \leftarrow R_{m0-15}$; LR ← retAddr	BLX Rm	2	-	-	-	-	0	1	0	0	0	1	1	1	1		Rm			0	0	0
Load from Literal Pool	Load PC-relative	Rd ← <label>	LDR Rd, <label>	2 or 1 (2)	-	-	-	-	0	1	0	0	1	Rd			imm8								
Load/store single data item	Word, register offset	Rd → [Rn + Rm]	STR Rd, [Rn, Rm]	2 or 1 (2)	-	-	-	-	0	1	0	1	0	0	0	Rm			Rn			Rd		
	Halfword, register offset	Rd → [Rn + Rm]	STRH Rd, [Rn, Rm]	2 or 1 (2)	-	-	-	-	0	1	0	1	0	0	1	Rm			Rn			Rd		
	Byte, register offset	Rd → [Rn + Rm]	STRB Rd, [Rn, Rm]	2 or 1 (2)	-	-	-	-	0	1	0	1	0	1	0	Rm			Rn			Rd		
	Signed byte, register offset	Rd ← [Rn + Rm]	LDRSB Rd, [Rn, Rm]	2 or 1 (2)	-	-	-	-	0	1	0	1	0	1	1	Rm			Rn			Rd		
	Word, register offset	Rd ← [Rn + Rm]	LDR Rd, [Rn, Rm]	2 or 1 (2)	-	-	-	-	0	1	0	1	1	0	0	Rm			Rn			Rd		
	Halfword, register offset	Rd ← [Rn + Rm]	LDRH Rd, [Rn, Rm]	2 or 1 (2)	-	-	-	-	0	1	0	1	1	0	1	Rm			Rn			Rd		
	Byte, register offset	Rd ← [Rn + Rm]	LDRB Rd, [Rn, Rm]	2 or 1 (2)	-	-	-	-	0	1	0	1	1	1	0	Rm			Rn			Rd		
	Signed halfword, register offset	Rd ← [Rn + Rm]	LDRSH Rd, [Rn, Rm]	2 or 1 (2)	-	-	-	-	0	1	0	1	1	1	1	Rm			Rn			Rd		

Divisione signed / Divisione Usigned

$\cdot 2^n$ / $:2^n$

Rd + Rn + Carry → Rd = Rn + carry

Reseta un bit (bit a 1 in R_M, in R_D verranno resettati a 0)

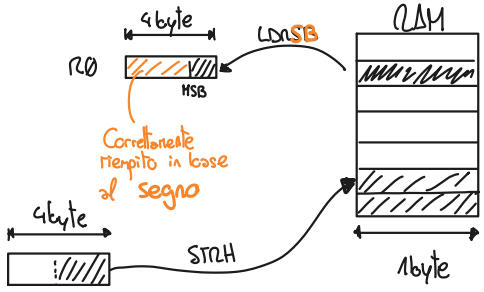
$R_n + (-R_M)$

(1) Depends on multiplier implementation

(2) 2 if to AHB interface or SCS, 1 if to single-cycle I/O port.

(3) N is the number of elements in the list

(4) N is the number of elements in the list including PC or LR.



Loadflags

{type}	Data type	Meaning	
	32-bit word	0..4'294'967'295 or	-2'147'483'648..+2'147'483'647
B	Unsigned 8-bit byte	0..255	Zero pad to 32 its on load
SB	Signed 8-bit byte	-128..+127	Sign extend to 32 bits o load
H	Unsigned 16-bit halfword	0..65'535	Zero pad to 32 its on load
SH	Signed 16-bit halfword	-32'768..32'767	Sign extend to 32 bits o load

Per store non importa più il valore viene semplicemente tagliato

Lars Kamm, Jan Wendler

OST Ostschweizer Fachhochschule

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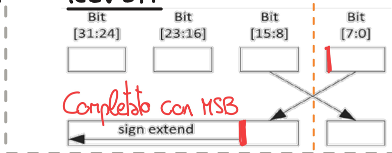
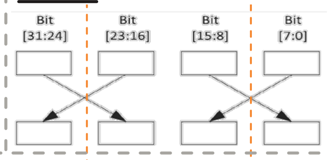
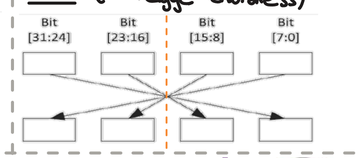
```
; Esempio di estensione R0 = 0x55AA8765

SXTB r1, r0    ; signed extend byte       -> r1 = 0x00000065
SXTH r1, r0    ; signed extend half-word -> r1 = 0xFFFF8765
UXTB r1, r0    ; unsigned extend byte     -> r1 = 0x00000065
UXTH r1, r0    ; unsigned extend half-word -> r1 = 0x00008765
```

**







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Group	Description	Operation	Assembler	Cycles		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Load/store single data item	Word, immediate offset	Rd → [Rn + #imm5]	STR Rd, [Rn, #<imm>]	2 or 1 (2)	- - - -	0	1	1	0	0	imm5					Rn			Rd		
	Word, immediate offset	Rd ← [Rn + #imm5]	LDR Rd, [Rn, #<imm>]	2 or 1 (2)	- - - -	0	1	1	0	1	imm5					Rn			Rd		
	Byte, immediate offset	Rd → [Rn + #imm5]	STRB Rd, [Rn, #<imm>]	2 or 1 (2)	- - - -	0	1	1	1	0	imm5					Rn			Rd		
	Byte, immediate offset	Rd ← [Rn + #imm5]	LDRB Rd, [Rn, #<imm>]	2 or 1 (2)	- - - -	0	1	1	1	1	imm5					Rn			Rd		
	Halfword, immediate offset	Rd → [Rn + #imm5]	STRH Rd, [Rn, #<imm>]	2 or 1 (2)	- - - -	1	0	0	0	0	imm5					Rn			Rd		
	Halfword, immediate offset	Rd ← [Rn + #imm5]	LDRH Rd, [Rn, #<imm>]	2 or 1 (2)	- - - -	1	0	0	0	1	imm5					Rn			Rd		
	SP-relative	Rd → [SP + #imm8]	STR Rd, [SP, #<imm>]	2 or 1 (2)	- - - -	1	0	0	1	0	Rd			imm8							
	SP-relative	Rd ← [SP + #imm8]	LDR Rd, [SP, #<imm>]	2 or 1 (2)	- - - -	1	0	0	1	1	Rd			imm8							
Generate PC-relative address	Form address from PC	Rd ← addr[<label>]	ADR Rd, <label>	1	- - - -	1	0	1	0	0	Rd			imm8							
Generate SP-relative address	Form address from SP	Rd ← SP + #imm8	ADD Rd, SP, #<imm>	1	- - - -	1	0	1	0	1	Rd			imm8							
Miscellaneous 16-bit instructions	Add Immediate to SP	SP ← SP + #imm7	ADD SP, #<imm>	1	- - - -	1	0	1	1	0	0	0	0	0	imm7						
	Sub Immediate from SP	SP ← SP - #imm7	SUB SP, #<imm>	1	- - - -	1	0	1	1	0	0	0	0	1	imm7						
* { Estende con MSB {	Extend signed halfword to word	Rd_{32Bit} ← signed Rm_{16Bit}	SXTH Rd, Rm	1	- - - -	1	0	1	1	0	0	1	0	0	0	Rm			Rd		
	Extend signed byte to word	Rd_{32Bit} ← signed Rm_{8Bit}	SXTB Rd, Rm	1	- - - -	1	0	1	1	0	0	1	0	0	1	Rm			Rd		
{ Estende con 0 {	Extend unsigned halfword	Rd_{32Bit} ← Rm_{16Bit}	UXTH Rd, Rm	1	- - - -	1	0	1	1	0	0	1	0	1	0	Rm			Rd		
	Extend unsigned byte	Rd_{32Bit} ← Rm_{8Bit}	UXTB Rd, Rm	1	- - - -	1	0	1	1	0	0	1	0	1	1	Rm			Rd		
	Push	regs → mem[SP--]	PUSH {<LoReglist>} *	1+N (4)	- - - -	1	0	1	1	0	1	0	0	lo_register_list							
	Push with link register	regs, LR → mem[SP--]	PUSH {<LoReglist>, LR}	1+N (3)	- - - -	1	0	1	1	0	1	0	1	lo_register_list							
	Disable interrupts		CPSID i	1	- - - -	1	0	1	1	0	1	1	0	0	1	1	0	0	0	1	0
	Enable interrupts		CPSIE i	1	- - - -	1	0	1	1	0	1	1	0	0	1	1	1	0	0	1	0
** {	Bytes in word	Rd ← RevByteOrder[Rm]	REV Rd, Rm	1	- - - -	1	0	1	1	1	0	1	0	0	0	Rm			Rd		
	Bytes in both halfwords	Rd ← RevHalfWords[Rm]	REV16 Rd, Rm	1	- - - -	1	0	1	1	1	0	1	0	0	1	Rm			Rd		
	Signed bottom half word	Rd ← RevLoHalf[Rm]	REVSH Rd, Rm	1	- - - -	1	0	1	1	1	0	1	0	1	1	Rm			Rd		
* Registro a SX sarà all'indirizzo più basso dello stack (salvo da DX a SX)	Pop	regs ← mem[SP++]	POP {<LoReglist>} *	1+N (3)	- - - -	1	0	1	1	1	1	0	0	lo_register_list							
	Pop and return	regs, PC ← mem[SP++]	POP {<LoReglist>, PC}	3+N (4)	- - - -	1	0	1	1	1	1	0	1	lo_register_list							
	Breakpoint		BKPT #<imm>	- (6)	- - - -	1	0	1	1	1	1	1	0	imm8							
	Send event		SEV	1	- - - -	1	0	1	1	1	1	1	1	0	1	0	0	0	0	0	0
	Wait for event		WFE	2 (7)	- - - -	1	0	1	1	1	1	1	1	0	0	1	0	0	0	0	0
	Wait for interrupt		WFI	2 (7)	- - - -	1	0	1	1	1	1	1	1	0	0	1	1	0	0	0	0
	Yield		YIELD	1 (8)	- - - -	1	0	1	1	1	1	1	1	0	0	0	1	0	0	0	0
	No operation		NOP	1	- - - -	1	0	1	1	1	1	1	1	0	0	0	0	0	0	0	0
Store multiple registers	Store Multiple	regs → mem[Rn++]	STM Rn!, {<LoReglist>}	1+N (3)	- - - -	1	1	0	0	0	Rn			lo_register_list							
Load multiple registers	Load Multiple, excluding base	regs ← mem[Rn++]	LDM Rn!, {<LoReglist>}	1+N (3)	- - - -	1	1	0	0	1	Rn			lo_register_list							
	Load Multiple, including base	regs ← mem[Rn]	LDM Rn, {<LoReglist>}	1+N (3)	- - - -	1	1	0	0	1	Rn			lo_register_list							
Conditional branch, and	Conditional (BEQ, ...)	if <cc> then PC ← label	B<cc> <label>	1 or 2 (5)	- - - -	1	1	0	1	cond				imm8							
Supervisor Call	Supervisor Call		SVC #<imm>	- (6)	- - - -	1	1	0	1	1	1	1	1	imm8							
Unconditional Branch	Unconditional	PC ← label	B<label>	2	- - - -	1	1	1	0	0	imm11										
Branch & Link 32Bit	With link	[15 - 0]	BL<label>	3	- - - -	1	1	1	1	0	S	imm10									
		[31 - 16]			- - - -	1	1	J1	1	J2	imm11										
Write and Read Special Register 32 Bit	Read special register	[15 - 0]	MRS Rd, <specreg>	3	- - - -	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	1
		[31 - 16]			- - - -	1	0	0	0	Rd				SYSm							
	Write special register	[15 - 0]	MSR <specreg>, Rn	3	- - - -	1	1	1	1	0	0	1	1	1	0	0	0	Rn			
		[31 - 16]			- - - -	1	0	0	0	1	0	0	0	SYSm							
Synchronisation 32 Bit	Instruction synchronization barrier	[15 - 0]	ISB	3	- - - -	1	1	1	1	0	0	1	1	1	0	1	1	1	1	1	1
		[31 - 16]			- - - -	1	0	0	0	1	1	1	1	0	1	1	0	option			
	Data memory barrier	[15 - 0]	DMB	3	- - - -	1	1	1	1	0	0	1	1	1	0	1	1	1	1	1	1
		[31 - 16]			- - - -	1	0	0	0	1	1	1	1	0	1	0	1	option			
	Data synchronization barrier	[15 - 0]	DSB	3	- - - -	1	1	1	1	0	0	1	1	1	0	1	1	1	1	1	1
		[31 - 16]			- - - -	1	0	0	0	1	1	1	1	0	1	0	0	option			

(Ogni bit a 1 rappr. registro (Da più piccolo a più alto)) → lo_register_list

⚠ (→ = store)

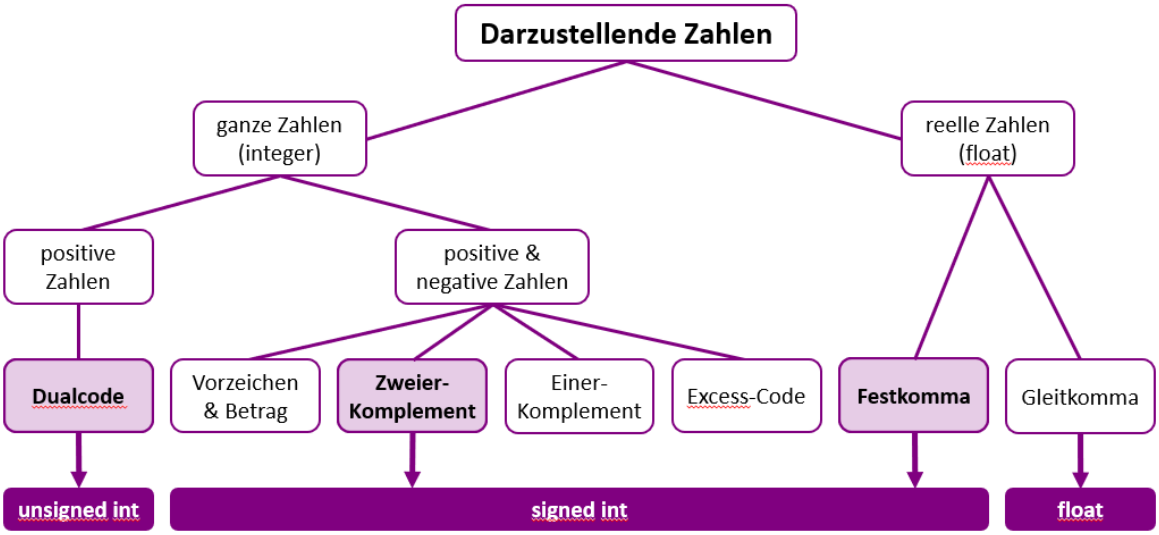
(5) 2 if taken, 1 if not-take
(6) Cycle count depends on processor and debug configuration.
(7) Excludes time spent waiting for an interrupt or event.
(8) Executes as NOP.

Mehrfache Datentransfers

- ! serve per aggiornare l'indirizzo Rn (dopo punterà a primo spazio libero dopo array)
- STM solo versione con !
- R⊕ bassi salvati a memory adress ⊕ bassi

- BL → salta all'indirizzo specificato e salva istruzione dopo attuale in LR (R14) per poter tornare indietro a fine fn
- BX Rn → Salta all'indirizzo contenuto in un registro (solitamente LR)
- CMN → Viene usato quando il secondo termine è negativo (a == -5 / a > -3) — Immediate verranno salvati su reg.

4 ZAHLENSYSTEME



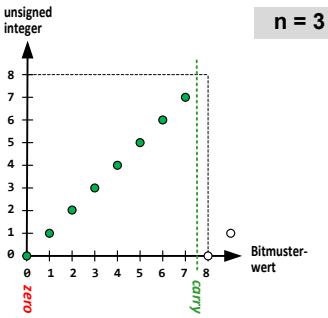
4.1 Vorzeichenlose Zahlen

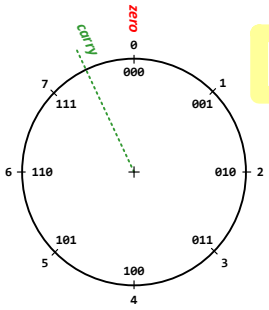
Dualcode (unsigned int)

$$W = \sum_{i=0}^{n-1} d_i \cdot 2^i$$

$$d_{n-1} \quad d_{n-2} \ldots \quad d_2 \quad d_1 \quad d_0$$

Relevante Flags: Zero, Carry/Borrow





Bereich
$0 \ldots 2^n - 1$

4.2 Vorzeichenbehaftete Zahlen

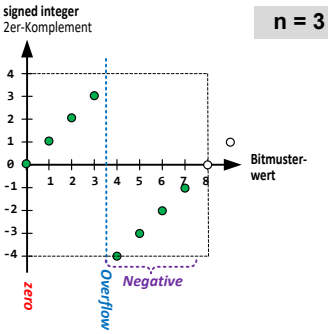
Zweierkomplement (signed int) → Comportamento ALU ∈ unsigned

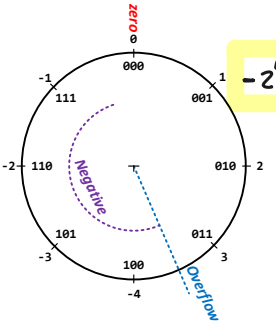
$$W = \left(\sum_{i=0}^{n-2} d_i \cdot 2^i\right) - d_{n-1} \cdot 2^{n-1}$$

$$d_{n-1} \quad d_{n-2} \ldots \quad d_2 \quad d_1 \quad d_0$$

Relevante Flag: Zero, Negativ, Overflow

MSB = 1





Bereich
$-2^{n-1}, \ldots, 2^{n-1} - 1$

4.3 Zweierkomplement-Negierung

Vorgehensweise:

1) Vorzeichenbehaftete Zahl in Binärcode darstellen
2) Alle Bits einzeln invertieren
3) Binäre Addition mit dem Wert +1
4) Resultat entspricht der negierten Vorzeichenbehaftete Zahl

Beispiel: -3_{10}=	**101_2**
Invertieren:	010_2
+1:	001_2
	011_2
Resultat:	$+3_{10}$

$2^{-1} = 0{,}5$; $2^{-2} = 0{,}25$ $2^{-3} = 0{,}125$; $2^{-4} = 0{,}0625$; $2^{-5} = 0{,}03125$

$2^{-6} = 0{,}015625$

$-d_n \cdot 2^n + d_{n-1} \cdot 2^{n-1} + \ldots d_0 \cdot n^0$

4.4 Festkomma-Zahlen

Bisogna correggere i numeri (Nearest) ! → Da Dec. 10 a Bin. 0b calcolare 1 passo in ⊕ nella parte dopo virgola e vedere se serve rundung

Vorzeichenlose Fixed-Point-Zahlen

Allgemeine Formel (Dualcode als Basis)

$$W = \sum_{i=-m}^{n-1} d_i \cdot 2^i$$

$d_{n-1}\ d_{n-2}\ \dots\ d_1\ d_0\ ,\ d_{-1}\ d_{-2}\ d_{-m}$

Wertebereich: $0.0 \dots 2^n - 2^{-m}$

Schrittweite: $\Delta X = 2^{-m}$ (Auflösung)

Max. Diskretisierungsfehler: $\frac{\Delta X}{2} = 2^{-(m+1)}$ (Quantisierungsfehler)

Vorzeichenbehaftete Fixed-Point-Zahlen

Allgemeine Formel (2er-Komplement als Basis)

$$W = \left(\sum_{i=-m}^{n-2} d_i \cdot 2^i\right) - d_{n-1} \cdot 2^{n-1}$$

$d_{n-1}\ d_{n-2}\ \dots\ d_1\ d_0\ ,\ d_{-1}\ d_{-2}\ d_{-m}$

Wertebereich: $-2^{n-1} \dots 2^{n-1} - 2^{-m}$

Schrittweite: $\Delta X = 2^{-m}$

Max. Diskretisierungsfehler: $\frac{\Delta X}{2} = 2^{-(m+1)}$

(La virgola è virtuale → Da bin. a Hex ignorarla)

4.5 IQ-Zahlenformate

Immer signed !

Integer/Quotient-Darstellung

Numero cifre decin. — Numero cifre dopo la virgola

Signed Fixed-Point mit n = I und m = Q.

$$W = -d_{I-1} \cdot 2^{I-1} + \sum_{j=0}^{I-2} d_j \cdot 2^j + \sum_{i=1}^{Q} d_{-i} \cdot 2^{-i}$$

$d_{I-1}\ d_{I-2}\ \dots\ d_1\ d_0\ .\ d_{-1}\ d_{-2}\ d_{-Q}$

Wertek. $-2^{I-1} \dots 2^{I-1} - 2^{-Q}$

Fraktionale Zahlen (oder Spezialfall I1Qx)

$$W = -d_0 + \sum_{i=1}^{Q} d_{-i} \cdot 2^{-i}$$

$d_0\ .\ d_{-1}\ d_{-2}\ \dots\ d_{-(Q)}$

Wertebereich: $-1.0 \le W \le +1.0 - 2^{-Q}$

Vorzeichen: Koeffizient d_0

Genauigkeit: $2^{-(Q)}$

⊕ Overflow solo quando eseguo −1 * −1 (Weniger aufwand)

4.6 Zahlenkreise

Addition: Zahlenvektoren hintereinander reihen (Anfang auf Ende) — A+B

Subtraktion: Zahlenvektoren stumpf gegeneinanderstellen (Ende auf Ende) — A−B

Bsp. 0x2 − 0x8

Unsigned Integer

Carry usato per 2ercomp. $A + \overline{B} + 1$

Relevante Flags: Zero, Carry/Borrow

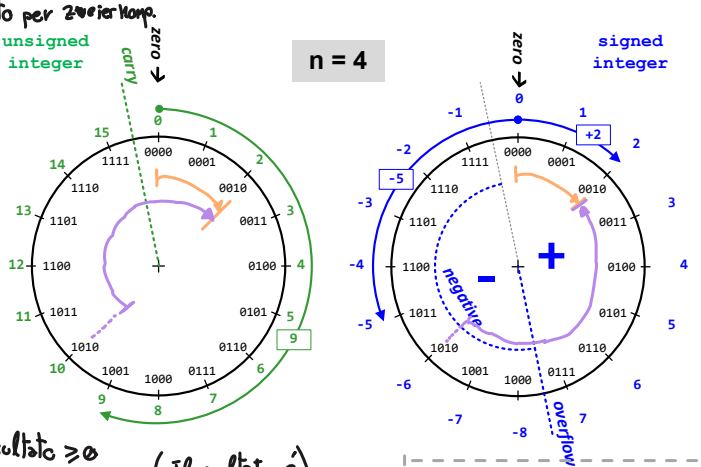
Zahlenvektoren immer Uhrzeigersinn

Signed Integer

(Primo bit 1) $(C_n \neq C_{n-1})$ (0 ... 0)

Relevante Flags: Negative, Overflow, Zero

positiv → Zahlenvektor Uhrzeigersinn
negativ → Zahlenvektor Gegenuhrzeiger



□ = Corrispondente carry nella sottrazione (Comportamento invertito rispetto a Σ) { 1 = Risultato ≥ 0 ; 0 = Risultato < 0 } (Il risultato è cmq corretto)

Nella ruota UNSIGNED tutti i vettori vanno in senso orario

Nella ruota SIGNED i vettori ≥ 0 in senso orario, $\vec{v} < 0$ in senso antiorario (regola ⇄ rimane)

Carry

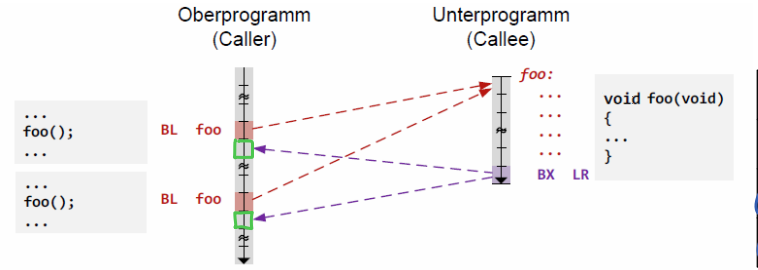
Addizione: se $C_n == 1$ → Carry = 1

Sottrazione: se $C_n == 0$ → Carry = 1 (Negato)

Bsp 0x2 − 0x8 → 0010 − 1000 = 1010 ⇒ Z = 0, C = 0, N = 1, V = 1

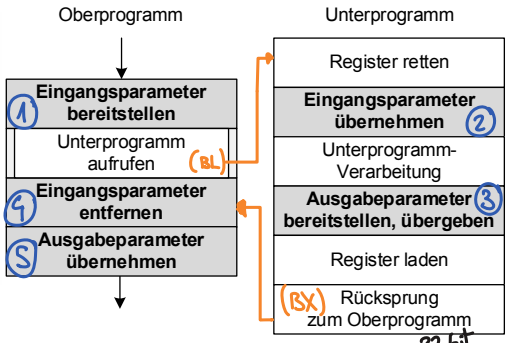
* Il compilatore può aggiungere allo stack frame LR e ulteriori registri per il calcolo, se necessario

5 SUBROUTINEN

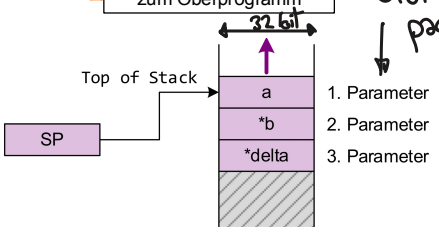


* Address istruzione salvato in LR

4) Se sono sullo stack, eliminare stack frame



Ordine param.



* Regeln zum ARM Architecture Procedure Call Standard (AAPCS)

- Die Register **R0-R3** werden als Parameter verwendet.
- Weitere Parameter werden auf den **Stack** gelegt. (> 4 Parametri)
- Der Rückgabewert (≤32Bit) wird in Register **R0** übertragen.
- Ein 64Bit-Rückgabewert wird in den Registern **R0** und **R1** übertragen.
- Funktionen müssen die Registerinhalte **R4-R11** während der Ausführung sichern und rekonstruieren.
- Die Register **R0-R3** und **R12** dürf(t)en in der Subroutine ohne Sicherung verändert werden.
- Es wird stets ein **8-Byte Alignement** auf dem Stack eingehalten.

Tutti i parametri (≤32 byte) vengono sempre passati come se fossero di 32 bit (int8_t → 32 bit)

Register	Verwendungszweck	Scratch	Sichern
R0	Erster Funktionsparameter, Return-Wert (8-/16-/32-Bit)	Ja	
R1	Zweiter Funktionsparameter, Upper Word Return-Wert (64-Bit)	Ja	
R2, R3	Dritter und vierter Funktions-parameter	Ja	
R4-R7	Registervariablen (temporär)	Ja	Ja
R8-R11	Compilerabhängige Verwendung	Ja	Ja
R12	Funktionsinternes temporäres Speicherregister für Sprünge	Ja	
R13	Stack-Pointer (SP, Top-of-Stack)	Diese Register werden für spezielle Zwecke verwendet	
R14	Link Register (LR)		
R15	Programm Counter (PC)		

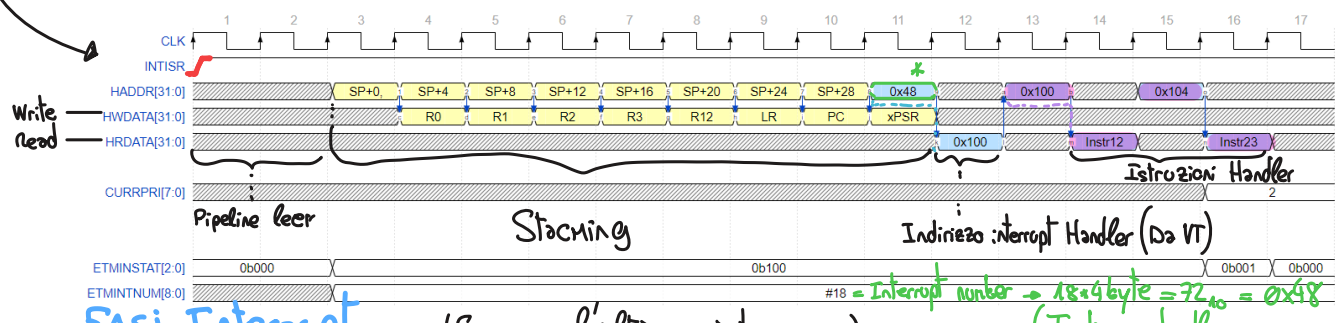
Low / High

Bus HADDR è sempre 1 ciclo in anticipo rispetto a H...DATA (Address Fetching)

6 INTERRUPTS & EXCEPTIONS

6.1 Vector Fetch & Stacking

□ = Registri salvati automaticamente



(Indirizzo handler in memoria, nella VT)

Fasi Interrupt

(Esegue l'ultima istruzione)

1. **Request:** (Arriva IRP) Pipeline «Execute» noch ausführen und der Pending Status setzen
2. **Stacking & Vector-Fetch:** Register auf Stack sichern und Sprung via Vektortabelle ausführen und Active Status setzen.
3. **ISR Handler:** Benutzercode ausführen und am Ende den Active Status zurücksetzen
4. **Unstacking:** Register wieder herstellen, Stack abbauen und Rücksprung mittels LR